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Heterogeneous Effects of Charter Schools: Unpacking Family Selection and Achievement Growth in Los Angeles

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ABSTRACT

Disparate findings on whether students attending charter schools outperform peers in traditional public schools (TPS) may stem from mixing differing types of charters or inadequately accounting for pupil background. To gauge prior family selection and heterogeneous effects, we distinguish between conversion and start-up charter schools, along with a third site-run model operating in the Los Angeles Unified School District (LAUSD). We find that TPS campuses converting to charter status (conversions) attracted more experienced and consistently credentialed teachers, and served relatively advantaged families, compared with newly created charter schools (start-ups), after tracking 66,000 students over 4 years, 2007–2011. Charters overall attracted pupils achieving at higher levels as they began a grade cycle (at baseline), relative to TPS peers, most pronounced among conversions that remained affiliated with the district. After matching students on their propensities to enter a charter school, we find that pupils attending charters outperformed TPS peers over the 4-year period. These benefits are most consistent and moderate in magnitude for middle school students. We observed significant though small effects in English language arts for pupils attending charter high schools. Latino students, mostly attending start-ups, enjoyed consistent benefits from attending a charter school.

KEYWORDS

achievement gains; charter schools; quasi-experimental methods

Charter schools continue to spread in many states, prompting the question of whether these independent, site-run organizations nurture steeper growth in children's learning, relative to peers attending traditional public schools (TPS). Research to date reveals widely varying achievement effects, differing among cities or states, and conditioned by the organizational features of schools (for reviews, Epple, Romano, & Zimmer, 2015; Fuller, 2015; Raymond, 2014). The search for factors that explain charter school effects is complicated by the difficulty of accounting for a priori effects on learning that emanate from pupils' family background, along with the varying capacity of charters to attract key resources, especially highly effective teachers.

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This article advances theory by examining differences in the pupils and teachers attracted to charters, relative to TPS, as well as variability between regular schools that convert to charter school status (*conversions*) versus newly created charters (*start-ups*). We describe how these organizational types may be located in differing communities and histories in the Los Angeles (LA) context. And we compare robust results from three estimation techniques: ordinary least-squares (OLS), conventional propensity-score matching, and recently developed genetic-matching techniques to estimate achievement effects from charter schools in the Los Angeles Unified School District (LAUSD).

We compare results when a prior-to-treatment achievement control is utilized, compared with estimating growth for students already enrolled in a charter school, and by grade level and cohort. This analytic strategy yields a more precise picture of the achievement benefits that stem from particular kinds of charter schools serving pupils at differing grade levels and over a varying count of years (dosage).

This analysis is based on tracking about 66,000 pupils in LAUSD, over the 2007–2011 period, the nation's second largest system, hosting 229 charter schools and 112 campuses that enjoyed site-level autonomy by the end of our time-series in 2011. We only include students with complete data over a 4-year period that commenced in Grades 2, 5, or 8, ensuring conservative controls for prior background. While our findings hold limited external validity, they do inform the selection processes and impact estimates of diverse charter schools in one major metropolitan area.

Why do estimates of charter school effects vary so widely?

National assessments find significant yet small advantages in learning curves for students attending charter schools, relative to peers in TPS (Raymond, 2014; for review, Epple et al., 2015). But comparative benefits vary widely across states and cities, and methodological worries persist that the estimation of benefits fails to properly account for the cognitive proficiencies or social skills of pupils prior to entering a charter, or bias introduced by unobserved confounders related to family attributes or practices that may explain differences.

Seeking clearer accounts for these varying effects, researchers have focused on how internal organizational features or external resources and regulatory environs act to mediate achievement effects (Furgeson et al., 2012; Hanushek, Kain, Rivkin, & Branch, 2007; Hoxby & Murarka, 2009; Zimmer, Gill, Booker, Lavertu, & Witte, 2012). We briefly review this literature on how organizational features of charters or prior student selection may help to explain varying estimates of charter school effects. We then describe our analytic strategy, distinguishing types of charters within LA, asking whether

conversion or start-up charters attract particular kinds of students, or draw differing teachers, compared with pupils and teachers hosted by TPS.

The diversity of charter schools within LAUSD must be taken into account. Independent charters—often called *start-ups*—do not have to abide by traditional attendance boundaries in admitting students. They do not hire teachers through the District's personnel process. Yet affiliated *conversion* charters respect attendance zones to varying degrees, at times filling seats with pupils from outside old attendance boundaries. Affiliated charters commonly work through the District's personnel system, although some gain flexibility to attract teachers that are preferred by the principal or committed to the mission of a particular school. The few conversions designated as *unaffiliated* generally operate like start-ups, as teachers sever their relationship with the District and LAUSD assumes minimal oversight.

The widening diversity of site-run schools

Policy makers and local school boards have authorized and funded an array of charter and similar, site-run schools over the past generation. This includes a widening diversity of organizational forms within the decentralized sector: most notably conversion and start-up charter schools, along with site-managed schools monitored by district offices, yet operating with charter-like autonomies. Nisar (2012) asked whether the degree of freedom from district regulation helped to account for variation in the effects of site-run schools on pupil achievement. Tracking pupil performance in 45 Milwaukee charters, 2005–2009, he found significant gains among start-up charters, but no discernible effects from pupils served by *instrumentality* charter schools, those that enjoy limited freedom from the district bureaucracy and labor rules.

Nasir inferred that the greater autonomy afforded to start-up charters, including flexible labor practices, acts to mediate stronger pupil growth, relative to charters still under the district's watchful eye (argued by Abdulkadiroğlu, Angrist, Dynarski, Kane, & Pathak, 2011, in a similar Boston study). It also appears that charter schools which have survived longer—better managing resource dependencies, enjoying stronger parental demand, or subjected to reauthorization processes—also exert modestly stronger benefits, relative to TPS (Raymond, 2014).¹

A second line of research asks whether *internal* organizational features of charters account for variation in achievement. Initial studies have found that smaller enrollments, longer instructional time, the mentoring of young teachers, and tighter collaboration help to explain the magnitude of learning gains (Furgeson et al., 2012; Zimmer & Buddin, 2006). One study of charters in New York City, tracking winners and losers of admission lotteries (close to an experimental design), found stronger achievement growth in schools

where principals observed and evaluated teachers more consistently and when they relied on “direct instruction” (Hoxby & Murarka, 2009). A smaller sample of 35 New York City charters studied by Dobbie and Fryer (2011) found that learning growth was steeper in charters when teachers analyzed student data, delivered tutoring, and expressed higher expectations for student learning (similar findings, Tuttle et al., 2013).

A third line of work finds heterogeneous effects of charters stemming from prior *student characteristics*. The national evaluation of charter middle schools, for instance, found no significant effects on student achievement or behavioral outcomes (other than stronger child and parent satisfaction with one’s school) for the mean pupil. But significant gains were observed for students from low-income families, based on tracking of admission lottery winners and losers (Gleason, Clark, Tuttle, & Dwyer, 2010). This suggests that prior student or home factors may set favorable conditions for positive charter school effects over time.

The lottery studies approximate a true experiment, although external validity is limited given that charter schools without waiting lists cannot be included. Still, this proved to be a small fraction of all New York City charter schools in Hoxby and Murarka (2009). And lottery studies ensure randomization between treatment and control groups. At the same time, studies drawing on large population samples of students moving into charters increasingly employ quasi-experimental techniques, including propensity-score matching.

Raymond (2014) estimated overall achievement for students attending an undifferentiated blend of charter schools in LA, relative to peers in traditional LAUSD schools after statistically matching students on observed characteristics over the period, 2009–2012. She found small test-score advantages enjoyed by charter students, relative to TPS peers, 0.07 SD higher in English language arts (ELA) and 0.11 SD higher in math on average. Latino pupils from low-income families experienced the strongest gains in charters, although the magnitudes of difference remained small. White and middle-class students in charters realized almost no difference in their rates of learning relative to TPS peers.

The strongest estimated achievement advantage felt by charter students occurred in middle schools, as reported by Raymond, the average difference reaching 0.22 SD for math, compared with TPS peers. Math scores ranged higher for pupils attending a charter for 3 years, rather than just 1 or 2 years. But no dosage effect was observed for ELA performance.

Our analysis moves beyond Raymond’s work by first distinguishing between conversion and start-up charter schools, then asking whether they draw-in differing kinds of students and families. Given their differing organizational histories in LA, we can’t assume that conversions and start-ups attract the same kinds of teachers or pupils, which in turn may affect their capacity to lift learning. We also delve deeper into Raymond’s finding that

middle schools may drive much of the overall charter advantage. The method employed by Raymond has been criticized for not controlling on students' academic proficiencies *before* entering a charter school ("pretreatment"), a fix that we build into our estimation techniques (Reardon, 2009).

Do differing charter schools attract particular students and teachers?

While these studies yield useful findings, they fail to conceptually situate charter schools and other site-run schools within the district-wide competition for preferred students and families, along with the institutional desire for stronger teachers. That is, we may see vivid forms of niche-seeking by charters, along with their competitors, as they pursue greater effectiveness within constraints (Fligstein & Dauter, 2007). These diversifying forms of schooling must attract sufficient resources *in relation to* other firms operating in their urban community. This involves competition for legitimacy, families, and material resources—especially teachers—factors they may condition each school's capacity to raise achievement.

Differing types of charter schools, for instance, may seek to protect the enrollment of—or seek—better performing students, or they may cater to families of certain ethnic or social classes. This may help to secure their niche, and hedge against resource uncertainties, in an otherwise competitive field of neighboring schools. While shaking free of central regulation and previous constraints on teacher hiring, charters can also actively seek certain kinds of students and families, rather than relying on the earlier attendance zones historically drawn by school districts.

Critics argue that better-off communities in LA, for instance, utilize chartering provisions to convert their TPS campus into a charter school, helping to seal-off enrollment demands from families outside the original neighborhood. LAUSD has granted deregulated freedom to schools in middle-class neighborhoods (of varying ethnic composition) under the Expanded School-based Management Model (ESBMM), which aims to mimic charter strategies. The ESBMM model offers a third type of school on which we focus, in addition to conversion and start-up charter schools. And conversion charters that remain legally affiliated with the District do so to collect larger amounts of federal Title I dollars. Such organizational behaviors may be legitimate; they are motivated by niche seeking, protecting or improving the flow of certain resources.

On the other hand, such niche seeking by strategic charter schools may act to harden the stratification of students, as differing families sort into particular schools. The *a priori* force of parents' social-class position is revealed in studies that track enrollment flows over time. Focusing solely on White families, researchers found that higher achieving children were more likely to seek out a charter option in four of seven major cities studied, compared

with lower achieving White peers (Zimmer et al., 2012). We similarly found that early TPS that converted to charter status in LAUSD (but not start-ups) tended to serve students achieving at higher levels at entry, relative to peers entering TPS (Lauen, Fuller, & Dauter, 2016).

Charters may also seek to attract particular kinds of teachers to pursue a specific curricular mission, nurture collaboration, or raise overall quality. Some charter schools rely heavily on “alternatively credentialed” teachers, such as Teach for America graduates. Start-up charters have typically employed younger, less costly teachers relative to TPS, both nationally (Bodine et al., 2008) and within LA (Fuller, Waite, Chao, & Benedicto, 2014).

Tracking who taught in North Carolina charter schools over a 13-year period ($n = 6,823$), Carruthers (2012) found that just over one fourth had migrated from TPS, displaying lower qualifications (graduate degrees, state licensing scores) and classroom effectiveness (estimated value-added scores), compared with TPS peers, especially those moving into predominantly Black charter schools.² Otherwise, little is known about how teachers migrate among TPS and differing types of charter schools.

One counter to this argument pertaining to organizational segmentation stems from findings showing that charters and similar site-run schools do not necessarily attract higher achieving students or particular kinds of teachers. Instead, some charter schools may become rather conventional over time—that is, isomorphic with TPS incumbents in their organizational field—in order to gain market credibility and attract more families (Huerta, 2009; Renzulli, Barr, & Paino, 2014).

Thus, another pivotal question emerges that’s highly relevant to the LA context: Do conversion or start-up charter schools draw in differing students, families, or teachers relative to TPS campuses?

Does organizational diversity stratify students and families?

We know that family demand expressed for charter schools can differ along lines of race, class, or home language. But evidence remains mixed on the extent to which differential selection into charter schools actually occurs. It appears that the extent to which “creaming” of stronger students takes place, even inadvertently, is conditioned by local dynamics: the mix of proximal options facing parents, demographic variety, and the practices of charter operators. What’s less clear is whether differing types of charter schools—such as conversions and start-ups—may be located geographically in ways that favor certain groups or families residing in prior attendance zones, and whether this necessarily leads to differing selection patterns.

Broad national samples consistently show that African American students and those from poor families are overrepresented in charter schools, relative to TPS enrollments, suggesting wide inclusivity among most charter

campuses (Epple et al., 2015; Raymond, 2013). Studying one metropolitan area, New York City, Hoxby and Murarka (2009) found few differences between students entering charter admission lotteries and TPS peers. Gauging the odds of selecting a charter school among previously lower or higher achieving students, Zimmer and colleagues (2012) found no differences for Blacks and Latinos, although higher achieving White students in their prior TPS were more likely to migrate to charter schools, compared with lower-achieving peers.

Selection may be conditioned by the form of choice that becomes available. Peterson, Campbell, and West (2002) found that Catholic and more religious parents, along with those with greater residential stability, were more likely to take up the offer of a portable school voucher. But applicants for New York City vouchers were more likely to be Black and receiving welfare benefits than nonapplicants (Howell, 2004). Tracking ethnically diverse students that entered Chicago's liberalized choice scheme, Lauen (2009) found that those from middle-class families on average traveled longer distances to private or competitive TPS, and enjoyed stronger learning gains—compared with peers from poor, mostly Black families who exited neighborhood schools at lower rates, traveled shorter distances, and displayed weaker gains when they did enter the wider market of schools. Focusing on predominately Black charters, it appears that less disadvantaged or better educated parents are more likely to select a charter school, relative to neighboring pupils who remain in TPS (Buckley & Schneider, 2005; Henig, Hula, Orr, & Pedescleaux, 2001).³

The geographical unit of analysis may help to explain differing findings on family selection. Nationwide, charters have become disproportionately located in lower-income urban centers, perhaps responding to where pent-up demand for alternatives is greatest. But among poorer neighborhoods, the count and variety of proximal school options appear to condition the individual family's propensity to enter the wider market, and over time this may shape local norms that encourage more parents to shop for higher quality schools (Dauter & Fuller, 2016).

Charter schools in some cases may manifest the prior structure of the local education field along lines of race and class, not exacerbate the stratification of families and schools. Renzulli (2006), for instance, found that a larger share of Black families enrolled their children in charter schools when living in a more racially segregated school district, compared with weaker demand for charters among Black parents who lived in more integrated districts.

The segmentation of schools along lines of race or social class may stratify the availability of key resources as well, including the quality of facilities, teachers, and learning materials. This concern, of course, has formed the basis of ample school finance litigation over the past half century (Rebell, 2009). Magnet, vocational, and the progressive “free schools” of the 1970s

contribute to the ancestry of site-run charter schools, each model pressed by certain groups or pedagogical advocates (Lubienski & Lubienski, 2006). Whether these segmented class or pedagogical preferences come to stratify pupils among charters vis-à-vis TPS remains an important empirical question. We pursue this topic in the case of LA.

The LA story: Do charter schools attract certain students and teachers?

LA exemplifies how urban school bureaucracies have come to accommodate, even compete with, the variety of school organizations that have sprouted over the past generation. This widening array of site-run, partially deregulated schools include conversion and start-up charter schools in the context of LAUSD, along with the similarly autonomous ESBMM campuses. The latter involves charter-like detachment from regulatory regimes and allows schools to operate under a “thin” labor contract, awarding principals greater authority to hire and fire teachers. Teachers in ESBMM schools remain within the district’s personnel and benefits system (Martínez & Quartz, 2012). While the United Teachers of LA opposes the spread of charter schools, it supports the ESBMM model as a viable competitor to charters.

One of LA’s earliest charter schools illustrates how family and educator interests, at times segmented along lines of race and social class, may correspond to patterns of family selection. Leaders and parents tied to Pacific Palisades High School, situated on the affluent west edge of the city, initially petitioned to become a district-affiliated (conversion) charter, affording limited autonomy from the downtown bureaucracy and labor constraints, while winning flexibility to select students residing as far east as the University of California, Los Angeles community.

Charter schools then spread across middle-class parts of West LA and the San Fernando Valley in the early 2000s, led by an early wave of conversion charters (A map of this growth, 2000–2015, is available from the authors.). Then, the LAUSD board began to approve charter applications from start-ups, including those designed by charter management groups, mostly situated in lower-income parts of downtown, East LA, and South LA (Kerchner, Menefee-Libey, Mulfinger, & Clayton, 2008; Ledwith, 2010).

These differing niches, carved out by distinct types of charter schools, stem from the local institutional history. The so-called Belmont Zone of Choice—created by LAUSD after agitation by Latino activists in the mostly immigrant Pico-Union area—abolished neighborhood attendance zones, invited charter companies to create new campuses, and sparked the eventual founding of 51 charter-like pilot schools (Fuller, 2010; Martínez & Quartz, 2012). At the same time, middle-class communities have used the ESBMM mechanism to win deregulation and site-run management, helping to sharpen their identity and hire preferred teachers.

What remains unknown is the extent to which these different types of site-run schools, still expanding across LAUSD, attract certain types of students and teachers. Then, in turn, how might differing inflows of families and resources act to condition the value-added achievement effects of conversion and start-up charters relative to TPS campuses?

Research questions: How diverse charter schools may yield differing effects

Given these persisting issues in the literature, we examined student selection and value-added effects of differing kinds of charter schools within the LA context. Our empirical analysis begins with two *descriptive questions*:

- RQ1. Do conversion and start-up charters, along with similarly site-run (ESBMM) schools and TPS, attract differing students in terms of ethnic background, language proficiency, social-class attributes, and achievement levels at baseline?
- RQ2. Do conversion and start-up charters, along with similarly site-run (ESBMM) schools and TPS, attract differing kinds of teachers in terms of ethnicity, credential levels, length of experience, and tenure status?

After discovering that distinct forms of charter and site-run schools do tend to serve particular students and teachers, at times differing from TPS peers, we then ask whether this conditions variable effects on student learning. This analysis is informed by the earlier research suggesting that start-up charters may show stronger achievement effects, since they enjoy greater independence, compared with conversion charters that remain partially entangled with district headquarters and negotiated labor rules.

We know little about whether particular kinds of pupils will benefit more from certain types of schools. For example, Angrist, Pathak, and Walters (2011) did find larger positive effects for Black and Latino pupils (who entered with lower test scores) after attending charter middle schools in Massachusetts, compared with White peers.⁴ And remember, we know that conversion charters initially operating in LAUSD tended to serve higher achieving students, then exerted few advantages, relative to TPS peers, net family background (Lauen, Fuller, & Dauter, 2015). So, to test for such heterogeneity of effects stemming from the differing forms of site-run schools, we focus on four *explanatory questions*:

- RQ3. Do students attending a charter school for 3 or 4 years outperform statistically matched peers in TPS that share similar attributes? Are these results sensitive to the method of estimation?

RQ4. Do charter achievement effects differ by school level, that is, among students attending elementary, middle, or high schools?

RQ5. Do charter achievement effects differ for students attending start-up versus conversion charter schools?

Method

Data

Longitudinal data obtained from LAUSD allowed us to track students attending a charter, ESBMM, or TPS over the 2007–2008 to 2010–2011 school years. These records include data on a variety of student attributes and test score results for ELA and mathematics in Grades 2 to 11. Data on charter students, including longitudinal test score information, were compiled by staff at the California Department of Education (CDE), along with teacher characteristics.

We then built tandem samples of students who attended a charter school for two or three consecutive years during Grades 3 to 5, Grades 6 to 8, or Grades 9 to 11, corresponding to the typical period of elementary, middle, and high school in LAUSD.⁵ Given the more recent founding of ESBMM schools, data were available for the two final years of the time series and included in descriptive analyses, but these schools were excluded from explanatory analyses.

The first sample includes pupils attending the same school throughout the designated periods and for whom a prior test score was available at Grades 2, 5, or 8. This limited the size of our student sample but ensured conservative controls on each child's prior proficiency level and family background. These sampled students are labeled *stayers*. They include 32,657 elementary (2,773 charter, 29,884 TPS), 1,018 middle (556 charter, 462 TPS), and 1,032 high school (214 charter, 818 TPS) students at baseline in 2007–2008. Comparison groups are TPS-enrolled students for each respective time period.

The second sample includes only students who moved from a TPS into a charter school, those we call *switchers*. This sample offers a more thorough control on overt bias since the key covariate (the baseline test score) is not contaminated by the child's treatment status (Ballou, Teasley, & Zeidner, 2008). On the other hand, this sample of switchers only includes students and families who considered moving to a new school prior to Grades 3, 6, or 9, a particular choice set, which constrains the generalizability of our estimates. This second sample included 2,677 elementary (174 charter, 2,503 TPS), 31,993 middle (1,848 charter, 30,145 TPS), and 28,315 high school (1,764 charter, 26,551 TPS) students at baseline in 2007–2008.

The resulting tandem samples of students resulted in those who attend entirely independent start-up charter schools, affiliated or independent conversion charters. The geographies in which conversion charters first

appeared in the District—generally but not always situated in economically better-off areas—plays a role in the demographic attributes of students and families as we detail below. The later spread of start-up charters—serving lower-income communities beginning in the mid-2000s—yields a differing demographic profile on average.

Measures

Achievement outcomes

California Standards Tests (CSTs) were administered to almost all students in the spring of each year in ELA and math between Grades 2 and 11. California had not vertically scaled these tests across grades (prior to adopting Common Core State Standards), so we calculated standard *z*-scores for each student's grade-level CST score. We did not estimate math achievement for high school students: these tests reflect multidimensional scales in math, including algebra, geometry, and calculus, and pupils complete these courses in different grades.

Student characteristics

Schools report various pupil attributes attached to CST test forms each year, including the child's ethnicity, designation as "limited English proficient," special education eligible, qualifying for subsidized lunch, and categories of parent education. The latter variable was missing for about one quarter of the student sample at baseline, mostly for young students (2007–2008). Given this covariate's possible importance as one predictor of selecting a charter school, we dropped students failing to report parental education. We reran analyses with imputed values for parent education, but this did not appreciably change our results (available from the authors).

Teacher characteristics

Viewing teachers as one key resource that charter and ESBMM schools attempt to attract, we obtained administrative data from the CDE to assess qualitative differences in teachers. Each LAUSD school reports annually on the count of teachers, ethnic affiliations of their staff, teachers with and without a full credential (as opposed to an "emergency credential" or student interns serving as instructors), whether tenured or not, years of teaching experience, and possessing a graduate degree beyond the bachelor's level.

Comparing results from three estimation methods

We utilize OLS regression and two quasi-experimental techniques to estimate the impacts of attending charter schools relative to TPS counterparts. We compare the resulting estimates for each subject (ELA and math) and school

level (elementary, middle, and high school). We also compare the results by *switchers* who attended for up to 2 years, and for *stayers* enrolled up to 4 years. Finally, we disaggregate charter students between those attending a start-up versus a conversion charter.

We begin with OLS regression as a conventional starting method. This approach assumes that an underlying linear form is the true model. This is naïve for drawing impact estimates using observational data, because OLS regression fails to account for the prior school-selection process or potential confounders that may affect the likelihood of selecting the treatment *and* shape achievement outcomes.

To address these shortcomings, we employed two types of matching techniques. Although this remains vulnerable to selection bias due to omitted variables, these techniques reduce the risk by matching students on observable covariates, avoiding strong assumptions about the functional form of covariates, while increasing the robustness of the treatment-effect estimates (Ho, Imai, King, & Stuart, 2007).

The core foundation for matching methods is Rubin's (1974) causal model, which conceptualizes causal inference in terms of the potential outcomes under treatment and control, only one of which is observed. The causal effect of charter school attendance can be defined as the difference between an observed outcome and its counterfactual (Holland, 1986; Rubin, 1974). In reality a variety of categorical (e.g., parent's education level) and continuous factors (e.g., prior achievement) influence student selection into a charter school. Finding exact matching pairs remains challenging.

Given the multiple factors that affect student selection of a charter school, nearest-neighbor matching is now often employed based on a propensity score. This technique has become common in the sociology or economics of education, due to its appealing theoretical properties (e.g., Berends, Goldring, Stein, & Cravens, 2010; Buckley & Schneider, 2005; Xiang & Tarasawa, 2015; Zimmer & Buddin, 2006a). Researchers assign a conditional probability of receiving a treatment to each individual student based on the relevant set of observed covariates. Matching on this propensity score will eliminate overt bias in the sample, but only if we know the correct propensity score or construct a full model to estimate it (Rosenbaum & Rubin, 1983). Because we do not know this propensity score, we must estimate it by including the potentially confounding covariates in a logistic or probit regression model.⁶

Genetic matching offers an alternative to propensity score matching by using the iterative machine-learning algorithm that matches individuals based on their weighted Mahalanobis distances in multivariate space (Sekhon & Mebane, 1998). Unlike propensity score matching that ties each treated unit to the nearest control unit on a unidimensional metric, genetic matching uses a generalization of the Mahalanobis distance metric (Diamond & Sekhon, 2013).⁷ Estimators based on the propensity scores (e.g., "nearest

neighbor” techniques) will provide consistent impact estimates only if the distribution of covariates conditioned on the estimated propensity score is the same for the treatment and the comparison groups.

While propensity score matching techniques does not necessarily guarantee that acceptable balance of covariates between treatment group and control group will be achieved, genetic matching typically achieves a satisfactory level of balance (Henderson & Chatfield, 2011). The genetic matching algorithm eliminates the need to manually and iteratively check the propensity score and uses a search algorithm to iteratively check and improve covariate balance. It is a multivariate matching method that uses an evolutionary search algorithm developed by Sekhon and Mebane (1998) to maximize the balance of observed covariates across matched treated and control units.

We report comparative estimates of treatment effects based on propensity score and genetic matching, expecting that genetic matching would provide less biased estimates. For propensity score matching, we fit probit regressions for obtaining propensity scores by specifying the baseline test score, special education status, limited English proficiency status, and categories of ethnicity and parent education. We use the same covariates for genetic matching. The Appendix details our identification strategies and the equations necessary for estimating propensity scores.

Throughout the analysis, we used the *matching* package in R software (R Development, 2013), offering a flexible tool for implementing a variety of algorithms, including propensity-score and genetic matching (Sekhon, 2011). For the sensitivity analysis, we used the *rbounds* package to conduct Rosenbaum’s method of sensitivity analyses (Keele, 2010).

Findings

Descriptive results: Differing flows of students and teachers (RQ1)

Our initial two research questions ask whether the various types of deregulated schools—start-up and conversion charters and ESBMM schools—differ in the kinds of students and teachers they attract. Tables 1 and 2 detail the attributes of pupils in the baseline year (2007–2008) for those attending elementary, middle, or high school. The far-right column shows *p*-values for the analysis of variance for each variable by type of school: TPS, start-up and conversion charters. Given the recent origin of ESBMM schools, appearing in 2009–2010, we considered this as the baseline year; complete data were available for just two of the nine ESBMM campuses.

We see that for *elementary schools* the conversion charters attracted pupils with considerably higher ELA and math scores, 0.26 SD and 0.26 SD higher at baseline, respectively, than the respective means for TPS peers. Baseline math scores for pupils attending start-up charters also ranged higher, 0.27 SD and 0.30

Table 1. Descriptive statistics for student attributes in the baseline year by type of school, 2007–2008 (Reported as percentages).

Grade level	Variables	Charter school			ESBMM ^a	p-value (ANOVA)
		Traditional public	Start-up	Conversion		
Elementary	Special needs designated	9.0	9.2	11.1		*
	Limited English proficient	49.9	32.2	40.9		***
	Subsidized lunch eligible	84.1	67.9	60.2		***
	Ethnicity	5.9	6.9	5.7		***
	Asian	77.1	51.9	65.0		***
	Latino	9.3	29.5	6.0		***
	African American	7.3	26.3	33.7		***
	White	0.3	0.7	0.3		***
	Other ethnicity	12.8	15.5	12.0		***
	Parental education	24.9	15.8	27.2		***
Middle	Less than HS diploma	31.8	46.5	53.5		***
	High school	11	8.9	10.5		***
	College or above	32.3	28.5	8.8		***
	Special needs designated	81.7	77.1	53.3		***
	Limited English proficient	6.6	4.1	10.2		***
	Subsidized lunch eligible	76.5	67	50.4		***
	Ethnicity	9.6	24.7	14.4		***
	Asian	6.9	15.1	40.4		***
	Latino	0.3	0.7	0.2		***
	African American	10.7	19.1	13.1		***
Parental education	Less than HS diploma	26.1	23.9	23.2		***
	High school	28.5	44.5	58.5		***
	College or above					***

(Continued)

Table 1. (Continued).

Grade level	Variables	Charter school			ESBMM ^a	p-value (ANOVA)
		Traditional public	Start-up	Conversion		
High	Special needs designated	10.5	6.6	10.1	9.3	***
	Limited English proficient	33	23.9	18.4	12.3	***
	Subsidized lunch eligible	76.6	68.3	54.2	63	***
	Ethnicity	5.4	4.2	10.7	20.7	***
		79.5	65	55.5	62.1	***
		9.2	16.8	14.5	12.4	***
		5.5	14.5	21.5	2	***
		0.3	0.1	0.5	2.8	***
	Parental education	8.6	16.2	14	16.3	***
		25.2	29.7	18.8	17	***
		24.5	37.7	44.6	34.4	***

^aExpanded School-based Management Model (ESBMM) schools with complete data began in 2009–2010, which becomes their baseline year.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 2. Sample counts and mean standardized test scores for students in the baseline year, 2007–2008.

Dataset	Variables	Traditional public	Charter school		ESBMM ^a	p-value (ANOVA)
			Start-up	Conversion		
Elementary	<i>n</i> students	48,051	2,267	1,818		
	<i>n</i> schools	435	46	16		
	ELA	−0.04	0.27	0.26		***
	Math	−0.04	0.30	0.26		***
Middle	<i>n</i> students	45,040	3,479	856		
	<i>n</i> schools	109	40	4		
	ELA	−0.04	0.24	0.8		***
	Math	−0.03	0.16	0.74		***
High	<i>n</i> students	49,171	4,697	3,999	1,099	
	<i>n</i> schools	60	42	10	2	
	ELA	−0.09	0.35	0.45	−0.06	***

Note. ELA = English Language Arts.

^aExpanded School-based Management Model (ESBMM) schools with complete data began in 2009–2010, which becomes their baseline year.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

SD greater in ELA and math than the respective means for TPS students. Conversion charters served a lower share of Latino pupils, compared with the mean TPS (65% versus 77%), and a much lower percentage of children eligible for subsidized lunches (60% versus 84%). In short, conversion charters fill niches in economically better-off parts of LAUSD. Differences were similar when comparing students among charter and TPS *middle schools*.

The organizational niches filled by start-up and conversion charters emerge even more vividly when turning to *high schools* and the two ESBMM high schools with complete data. The 49 start-up charter high schools included in our sample enrolled pupils with significantly higher test scores at baseline: the mean ELA score for these students was 0.35 SD higher than for TPS peers. Start-up charters served a lower share of Latino pupils than traditional schools.

Mean parental education also ranged higher for students in start-ups relative to TPS peers (38% versus 25% of parents with at least some college, respectively). The high school sample included four conversion charters, and they served much larger proportions of White students with better-educated parents than TPS peers. The 49 start-up high schools, in contrast, tended to serve low-income Latino students, closely resembling TPS peers. The ESBMM high schools enrolled a higher share of Asian students and a smaller share of pupils from poor families than TPS counterparts.

These sectors also varied in terms of the kinds of teachers that each attracted and retained. Many conversions essentially inherited their teaching staff after winning independent status, while gaining discretion to attract the preferred mix of new teachers in the future. Table 3, beginning with *elementary schools*, shows that start-up charters employed lower shares of tenured

Table 3. Descriptive statistics for teacher resources in the baseline year, 2007–2008 (Reported as percentages).

Grade level	Teacher attributes	Traditional public	Charter school		ESBMM ^a
			Start-up	Conversion	
Elementary	Full credential	97.2	79.8	94.3	
	Years of teaching ^b	12.2	4.8	10.0	
	Tenured	85.9	18.9	63.5	
	Highest degree	23.1	11.8	15.2	
		59.6	67.6	71.2	
	Doctorate	17.4	20.6	13.7	
	Baccalaureate	12.3	14.3	12.2	
	Ethnicity	38.7	28.4	30.7	
	Latino	10.7	15.0	3.8	
	African American	37.8	40.9	52.3	
Middle	White	0.5	1.5	1.0	
	Other ethnicity	90.3	65.6	94.2	
	Full credential	10.5	4.4	9.6	
	Years of teaching ^b	74.6	10.1	60.8	
	Tenured	20.8	9.4	19.4	
	Highest degree	53.6	64.8	64.1	
	Doctorate	25.6	25.9	16.5	
	Masters	12.7	12.9	16.9	
	Baccalaureate	29.3	27.2	25.3	
	Ethnicity	12.3	15.8	3.9	
High	Latino	44.9	43.2	53.2	
	African American	0.8	0.9	0.7	
	White	87.6	60.8	88.6	82.4
	Other ethnicity	10.2	4.8	10.5	9.2
	Full credential	69.5	11.2	60.8	64.0
	Years of teaching ^b	22.5	13.6	24.4	19.9
	Tenured	49.0	65.8	57.6	48.7
	Highest degree	28.5	20.6	18.1	31.4
	Doctorate	12.2	13.1	12.7	14.1
	Ethnicity	26.5	27.7	18.1	22.6
	Latino	12.5	13.4	8.4	13.2
	African American	48.1	45.1	60.2	49.1
	White	0.7	0.6	0.6	1.0
	Other ethnicity				

^aExpanded School-based Management Model (ESBMM) schools with complete data began in 2009–2010, which becomes their baseline year. ^bYears of teaching^a pertains to working in one's present and earlier schools.

teachers or those with full credentials, although charter elementary schools tended to employ a higher share of teachers with master's degrees, compared with TPS peers. Just 19% of elementary teachers at start-up charters enjoyed tenure at baseline, compared with 63% employed by conversion charters and 86% at TPS campuses. These differences are reflected in the mean years of teaching experience: 4.8 years for start-up teachers, and 10.0 and 12.2 years in conversion charters and TPS, respectively. Conversion elementary schools employed a higher share of White teachers and few Black teachers, compared with start-ups and TPS campuses.

Sector differences were similar at the *high school* level. Three fifths of all teachers reported White ethnicity at conversion high schools, compared with 48% at TPS and 45% at start-up campuses. And start-up charters relied more on young, less experienced teachers with master's degrees, compared with TPS or conversion charters.

Descriptive differences in “post-treatment” achievement (RQ2)

Table 4 reports achievement levels in the final year of the time-series (2010–2011). These findings detail the kinds of students that remain in each sector at the end of the 4-year tracking period. Note that pupil achievement levels are not yet adjusted for prior family background or matched propensities to enter a treatment.

At the *elementary school* level, we see that students attending charters outperformed their TPS peers by significant margins after experiencing these schools. Pupils attending start-ups scored 0.21 SD and 0.17 SD higher in ELA and math, respectively, compared with the mean TPS student. Elementary students attending a conversion charter showed stronger achievement without prior covariate controls: 0.41 SD higher in ELA and 0.30 SD in math, relative to the TPS means.

Table 4. Mean standardized test scores for students in the final year of time-series, 2010–2011.

Dataset	Variables	Traditional public	Charter		ESBMM	p-value (ANOVA)
			Start-up	Conversion		
Elementary	n students	43,595	3,419	1,809		
	n schools	442	65	15		
	ELA	−0.05	0.21	0.41		***
	Math	−0.03	0.17	0.30		***
Middle	n students	40,690	4,686	794		
	n schools	88	54	2		
	ELA	−0.04	0.28	0.66		***
	Math	−0.04	0.26	0.45		***
High	n students	30,182	4,965	2,776	907	
	n schools	81	49	8	2	
	ELA	−0.04	0.13	0.44	−0.13	***

Note. ESBMM = Expanded School-based Management Model; ELA = English language arts.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Students attending conversion charter *middle schools* outperformed TPS peers by 0.66 SD in ELA and 0.45 SD in math by the end of the tracking period. And start-up charter middle schools display stronger achievement than TPS peers by 0.28 SD in ELA and 0.26 SD in math. Sector achievement differed for *high school* pupils as well, except that those attending the two ESBMM schools performed at lower levels than TPS peers.

Estimating charter school effects by school level (RQs 3 and 4)

We summarize in Table 5, the effects of charter school attendance, estimated by three different identification strategies, by school grade level, and by subject after taking into account family background, and then prior achievement. The OLS analyses regress test scores on the student-level covariates (labeled *Reg I*), and the student-level fixed effects models for *switchers* (*Reg II*). Details of these OLS regression analyses are presented in the Appendix. Next, two matching estimates report the average treatment effect on the treated (ATT) using the student-level covariates. The propensity-score matching estimate is based on the propensity scores obtained from a probit regression that includes the covariates in Table 1 (*P-match*, see the Appendix). Genetic matching estimates report the ATT when matched via the genetic algorithm, using the same student-level covariate controls (*G-match*).

Given that G-match offers the least biased estimates among the three analytic strategies, we focus on these results. We do, however, show that results are sensitive to how thoroughly one controls on family background and baseline achievement levels. In addition, we highlight differences among grade levels and results for charter school stayers and switchers.

Beginning with pupil *switchers* into *elementary* charter schools, we estimated similar gains relative to their TPS peers in ELA and math—but results were not significant. In contrast, for the *middle school* switchers, charter students showed significantly higher test scores on average, by about 0.15 SD in ELA and 0.19 SD in math, compared with TPS peers. At the *high school* level, switchers showed significantly higher achievement differences in ELA relative to TPS counterparts at a small degree of magnitude, a 0.11 SD difference. In addition, charter effects in math were significant for middle school switchers and consistent across the different estimation methods.

Results for charter school *stayers* differed in several ways. At the *elementary* level, charter students displayed gains in ELA only, a 0.11 SD advantage vis-à-vis TPS peers. In addition, the estimates for stayers proved sensitive to the identification strategy. This is not surprising because the key covariate, the student's baseline score, may be contaminated by one's treatment status, although controlled for. Still, the P-match showed results that were quite similar to G-match results.

Middle school stayers demonstrated higher learning gains in ELA relative to TPS counterparts. Unlike the ELA findings, simple regression yielded

Table 5. Estimated effects of charter school attendance for “switchers” and “stayers.”

Grade level	Estimation method	Switchers				Stayers			
		ELA		Math		ELA		Math	
		Est.	SE	Est.	SE	Est.	SE	Est.	SE
Elementary	Reg I	0.06	0.05	0.00	0.06	0.12	0.02	0.01	0.02
	Reg II	−0.07	0.04	−0.09	0.04	−		−	
	P-match	0.04	0.08	−0.01	0.08	0.09	0.02	0.01	0.02
	G-match	0.04	0.07	0.02	0.07	0.11	0.02	0.01	0.02
Middle	Reg I	0.16	0.01	0.22	0.02	0.36	0.05	0.30	0.07
	Reg II	0.13	0.01	0.16	0.02	−		−	
	P-match	0.13	0.02	0.19	0.02	0.30	0.09	−0.10	0.15
	G-match	0.15	0.02	0.19	0.02	0.19	0.09	0.13	0.10
High	Reg I	0.12	0.01	0.19	0.02	−0.06	0.05	−	
	Reg II	0.19	0.01			−		−	
	P-match	0.18	0.04			0.02	0.07		
	G-match	0.11	0.01			−0.07	0.06		

Note. ELA = English language arts. Reg II for stayers are not applicable because the treatment flag is no longer time-varying covariate for stayers, and cannot be specified in the student-level fixed-effects model.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6. Heterogeneous effects of start-up and conversion charter schools for elementary school “switchers.”

	Start-up charters						Conversion charters					
	ELA			Math			ELA			Math		
	Est.	SE	N (tr)	Est.	SE	N (tr)	Est.	SE	N (tr)	Est.	SE	N (tr)
Reg I	0.00	0.06	158	−0.04	0.07	157	0.21 *	0.10	55	0.12	0.11	56
Reg II	−0.08	0.04	158	−0.13 **	0.05	157	−0.02	0.07	55	0.02	0.08	56
P-match	0.07	0.08	158	−0.06	0.08	157	0.23	0.15	55	0.11	0.15	56
G-match	0.01	0.08	158	−0.02	0.08	157	0.22	0.14	55	0.15	0.12	56

ELA = English language arts.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

significant charter school effects in math, but these advantages diminished when employing matching techniques. This implies that the estimated treatment effects from the parametric regression approach stems from selection bias.

In summary, students who *switched* from a TPS elementary school into a charter middle school outperformed peers who remained in a TPS middle school. These switchers displayed advantages of 0.15 SD in ELA and 0.19 SD in math on average. These differences should be interpreted as modest in magnitude yet significant. Elementary students who *stayed* in a charter school displayed small learning advantages relative to TPS peers in ELA only: an advantage of 0.11 SD on average. And middle schoolers who switched in charter high schools showed achievement advantages relative to TPS peers at a similar magnitude (0.11 SD). These magnitudes of difference are similar to estimates for Boston charters (Abdulkadiroğlu et al., 2011; Angrist et al., 2011), based on admission lotteries (ELA, 0.08 SD; math, 0.21 SD)

**Do charter effects vary between conversion and start-up organizations?
(RQ 5)**

Given the differing positioning of conversion and start-up charters—with regard to pupils and teachers selected—this may condition their varying capacity to raise student achievement. We saw how students switching into

Table 7. Heterogeneous effects of start-up and conversion charter schools for middle school “switchers.”

	Start-up						Conversion					
	ELA			Math			ELA			Math		
	Est.	SE	N (tr)	Est.	SE	N (tr)	Est.	SE	N (tr)	Est.	SE	N (tr)
Reg I	0.16 ***	0.01	1,683	0.26 ***	0.02	1,678	0.11 **	0.03	324	−0.08	0.05	322
Reg II	0.14 ***	0.01	1,683	0.19 ***	0.02	1,678	0.08 **	0.03	324	0.00	0.04	322
P-match	0.13 ***	0.02	1,683	0.23 ***	0.03	1,678	0.12 **	0.04	324	−0.05	0.05	322
G-match	0.17 ***	0.02	1,683	0.25 ***	0.03	1,678	0.11 **	0.04	324	−0.10	0.05	322

ELA = English language arts.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 8. Heterogeneous effects of start-up and conversion charter schools for high school “switchers.”

	Start-up				Conversion			
	ELA				ELA			
	Est.		SE	N (tr)	Est.		SE	N (tr)
Reg I	0.11	***	0.01	2,136	0.10	***	0.02	1,406
Reg II	0.23	***	0.10	2,136	0.10	***	0.03	1,406
P-match	0.10	***	0.02	2,136	0.10	***	0.03	1,406
G-match	0.12	***	0.02	2,136	0.09	***	0.02	1,406

ELA = English language arts. We do not have results for Math at high school level due to the multi-dimensional test scores in Math at high school level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

charter middle and high schools enjoyed learning advantages vis-à-vis TPS peers, while elementary school switchers did not.

Achievement advantages did not differ markedly between elementary switchers attending start-up versus conversion charters (Table 6). However, we did observe heterogeneous effects for switchers at the *middle school* level, particularly in math. Students switching into a *start-up* middle school displayed steeper learning curves: 0.17 SD in ELA and 0.25 SD in math, compared with the mean TPS peer (Table 7). The latter effect size reaches a moderate level. The corresponding estimates for pupils switching into *conversion* charters were 0.11 SD in ELA and an insignificant difference in math, compared with the mean TPS peer. Note that the descriptive statistics in Table 4 showed much higher outcomes for pupils attending conversion charters, compared with peers attending start-ups or TPS. But when matching techniques are employed to pair students with similar student-level background characteristics, charter school effects from conversions disappeared in math, while remaining at a similar level observed for peers attending start-up charters. Heterogeneous effects were not observed at the high school level (Table 8).

Latino students benefit most from charter schools

To further locate value-added effects of charter schools, we interacted conversion or start-up status with student ethnicity. Overall, Latino students in both subsectors displayed the most consistent and strongest effects in math or ELA at the middle or high school level. These stronger benefits for Latino youngsters are consistent with Raymond’s earlier findings, after we utilized a differing analytic method and longer time series. Our estimation models with interaction terms are detailed in the Appendix (Table 2).

African American students enjoyed significant benefits, relative to White peers, in ELA at middle school charters. Also notable is that benefits in math were enjoyed mostly by Latinos, even within conversion charters, where ethnic differences were minimal in terms of value-added benefits.

Checking for covariate balance after statistical matching

Diverging results between P-match and G-match techniques may stem from differing patterns of balance in the covariates. Good balance is important to ensure that the groups being compared offer sufficient counterfactual cases for one another, at least on attributes that we can observe (Ho et al., 2007; Rubin, 2005). Balance levels by the P-match and G-match techniques were similar overall, although genetic matching tended to show higher *p*-values when applying *t*-test or KS tests, indicating a more even balance between treatment and control groups.⁸ Thus, the charter school effects presented previously do not appear to result from differences between treated and untreated groups on the covariates used in this study.

One exception that shows a discrepancy between P-match and G-match arose for middle school stayers in ELA. To diagnose the source of this difference, we present the covariate balance achieved by two different matching methods in Figures 1 and 2. In these two graphs, the solid circles represent the *p*-values for *t*-test and KS tests, respectively, before matching on covariates; the solid triangles represent the *p*-values after matching. We

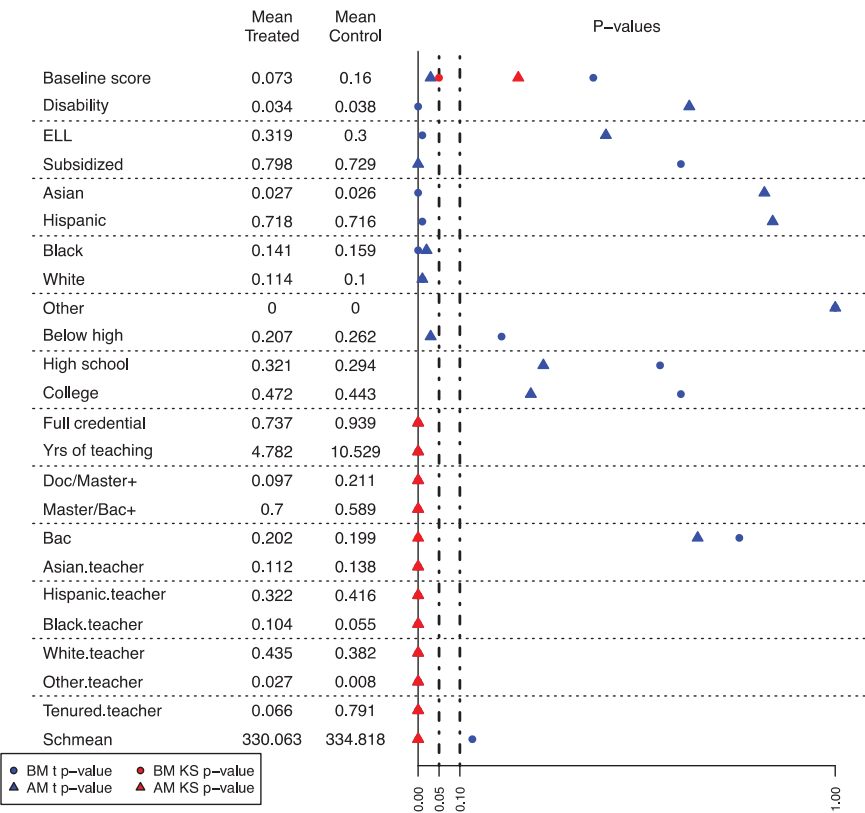


Figure 1. Covariate balance after propensity score matching for middle-school "stayers" illustrated with math achievement data.

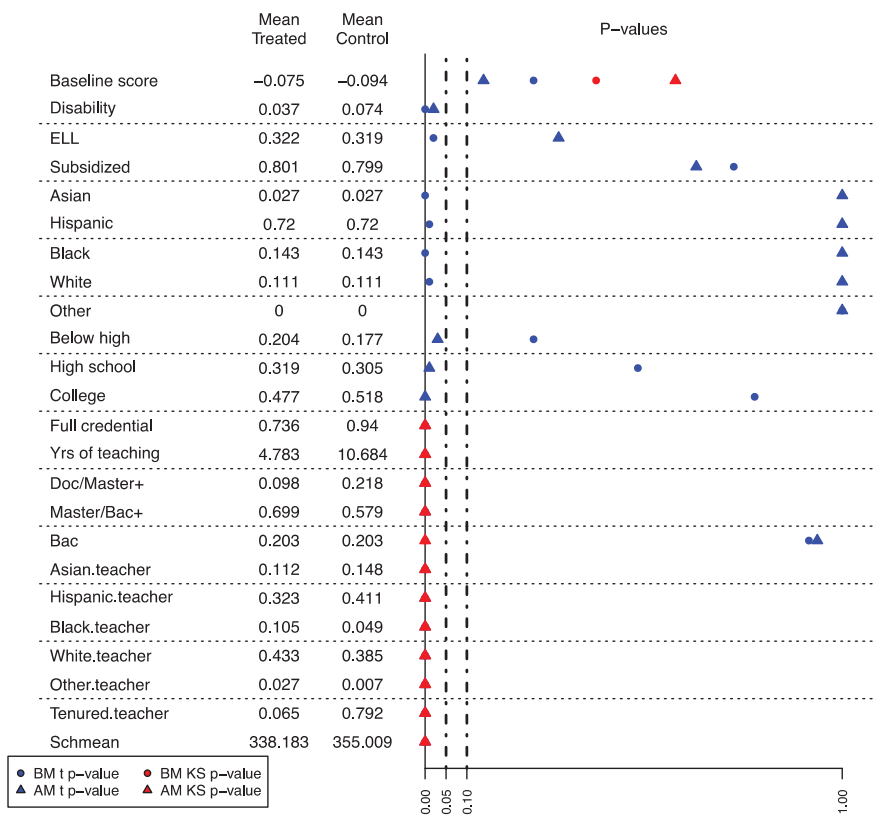


Figure 2. Covariate balance after genetic matching for middle-school “stayers” illustrated with math achievement data.

expect the dots to locate on the right side of the two dashed lines, indexed for p -values of less than 0.05 or 0.10, when each covariate is balanced between treatment and control groups.

By comparing [Figures 1 and 2](#), we clearly see that the key covariate, the student’s baseline test score, is not balanced after matching that relies solely on the propensity score. Yet the balance for other covariates, including the baseline score, improved when genetic matching was employed. This informs why we conclude that the genetic matching estimates are less biased than those derived from propensity score matching.

Practically speaking, unbalanced covariates may be more prevalent in organizational fields with highly differentiated firms. Checking the balance in covariates after attempts at matching yields further information about the distinctiveness of each organizational form, in our case indicated by sharply varying types of students and teachers. This is illustrated in [Figures 1 and 2](#), where both matching techniques failed to achieve balance for school-level covariates, starting with whether teachers were fully credentialed.

Sensitivity analysis

Overall, we observed achievement gains for middle- and high-school *switchers* relative to their respective TPS peers. These estimates may still be biased by unobserved covariates not included in the matching process. To address this concern we used Rosenbaum's method of sensitivity analyses (Keele, 2010), estimating how large the difference in the underlying probability of receiving treatment must be to alter the interpretation of estimates based on the two matching techniques.

That is, sensitivity tests assess if estimates are robust to bias due to remaining imbalances in any observed or excluded covariates after matching (Rosenbaum, 2002). The test then assesses the lower and upper bounds of a matching estimate when one observation in a matched pair is allowed to have a higher probability of receiving treatment due to observed or unobserved confounders. For example, when setting $\Gamma = 3$, one observation in a matched pair could be three times as likely to have received the treatment without eliminating the observed effect of that treatment. If the bounds include zero at low levels of Γ , then the estimate should be considered highly sensitive to selection bias.

Results for charter middle schools were most robust, largely immune to the potential effects of omitted selection factors (details appear in the Appendix). In our case the p -values of the effect for middle-school charter switchers after genetic matching is significant at the conventional level ($\alpha = 0.05$) until $\Gamma = 1.84$ for ELA and $\Gamma = 1.61$ for math. The corresponding values are 1.40 for elementary charter stayers for ELA performance, 1.49 for high-school charter switchers for ELA. This suggests that reported value-added effects of charter schools are not sensitive to unobserved covariates in the first-stage selection model.

Discussion and policy implications

We have detailed how differing types of charter schools and similar site-run campuses often attract particular students and teachers. One distinct theoretical lesson from these findings is that conversion charters may often be situated in economically better-off communities—depending on local demographics and policy histories—then draw higher achieving students at baseline, relative to both start-ups and TPS. Conversions, at least in the LA context, also drew more experienced and often White teachers, creating quite different conditions for learning relative to other schools. These differences in family selection and teacher resources may then condition the estimated value-added benefits of charter schools.

Overall, charters in LA served higher achieving students at baseline, then relied on variably experienced teachers. *Start-up* charters occupy a particular

position in this segmented organizational field and teacher workforce, relative to TPS. Mean years of experience for elementary teachers equaled 4.8 in start-up charters versus 12.2 years in TPS. In addition, *conversion* charters are distinctly positioned in LAUSD to pull-in higher achieving students and experienced teachers, in part given their historical positioning in economically better-off parts of the district.

Yet we should not think about “creaming” of higher achievement students in simple ways. The locational history of conversion charters in LA—where former TPS located in middle-class areas aggressively sought charter status—may better explain the “creaming” of abler students into conversions, not the aggressive recruitment of higher achievers from outside former attendance zones. Future work might further disaggregate those conversions that remain tied to the district personnel system, relative to those that operate entirely independent of the downtown bureaucracy. This offers another institutional distinction that may condition where conversions locate and the families they opt to serve.

Conversion high schools display a distinct complexion as well, where three fifths of teachers were White, compared with 48% at TPS and 45% at start-up charters. The racial or class-related positioning of charter schools, relative to TPS, emerges when unpacking student attributes as well. Conversion elementary schools serve pupils with higher ELA and math scores, 0.34 SD and 0.32 SD greater at baseline relative to TPS students, respectively. Even baseline math scores for pupils attending start-ups, more often of color and from less educated families, ranged a bit higher, 0.21 SD greater than for TPS peers. We saw how conversion charter elementaries served a much lower share of Latino pupils, compared with the mean TPS, and a much lower percentage of children eligible for subsidized lunches.⁹

We did observe that conversion charters effectively maintain baseline advantages for their relatively advantaged students, compared with start-ups and TPS. Yet charter middle schools then lift the learning trajectories of their students relative to peers who remain in TPS. These gains for charter middle schoolers are consistent across the three methods of estimation, although they vary in magnitude. The analysis for pupils switching into a charter school offers the best approximation of the discrete treatment impact. At the middle-school level, switchers into charters displayed significantly higher scores: about 0.15 SD higher in ELA and 0.19 SD higher in math, compared with TPS peers.

Similarly, elementary-level stayers attending charters displayed small achievement advantages, higher scores in ELA (0.09 SD) and no advantage in math, compared with TPS peers. And we detected small yet significant benefits for pupils that switch into a charter high school from a TPS (0.11 SD). Perhaps most encouraging across grade levels and subject areas is the finding that Latino pupils benefit most consistently and at stronger levels of

magnitude, relative to peers attending TPS. *How* charters better serve this pivotal population of kids is a question that deserves more research.

While conversion charters effectively maintain or widen differences in student performance vis-à-vis TPS peers, start-ups held slight, yet at times significant, benefits after taking into account prior achievement and family background. One notable exception is start-up middle schools, which significantly boost math achievement above levels observed among TPS peers. And start-up charters appear to benefit many elementary and high school-level children from low-income families at least in ELA and at significant levels of magnitude. Overall, the organizational position held by start-up charters may result in drawing less effective teachers or resources—including differing populations of families, compared with those drawn by conversion charters.

These results apply only to the local dynamics of LA, one metropolitan area with a growing range of school options, even across lower-income communities. The external validity of these results could be widened by estimating differential selection outcomes and impact estimates of start-up and conversion charters in other cities. Ideally, students would be tracked for longer periods of time. Our design would have been stronger if pupil achievement trends could have been tracked at least 3 years prior to switching into a charter school to better isolate the treatment effect. A proficiency control measure prior to school entry could lead to less biased estimates of the elementary charter treatment effect. This would require more thorough and uniform data collection by charter school authorities.

Future studies might track students in noncharter options as well. Our data were insufficient to carefully compare the ESBMM schools with charter or TSP performance. Magnet, private, and other options would ideally be included in future student tracking studies. And thicker measures of home practices and the variable press of parents on their children to do well in school would reduce potential bias stemming from unobserved covariates (beyond conventional demographic characteristics) in the first-stage propensity model.

Future research might examine how this positioning of conversion and start-up charter schools—in terms of geography and the social class of families served—conditions the learning trajectories of students. Qualitative fieldwork could inform how diverse charters at times select particular families, then work to lift achievement. Conversions may also locate in areas with a diverse range of families and pupil achievement levels, aiming to attract families disaffected by low-quality schools in their local field (suggested by Bilfulco & Buerger, 2012). This holds implications for the proximal choice sets that poor or middle-class parents consider.

Policy makers should consider whether the spread of start-up and conversion charters may further separate high- from low-achieving students across LAUSD, organizational diversity that may worsen segregation, even inadvertently.

Nor do we understand how this evolving landscape of alternative schools may harm the educational trajectories of weaker students who remain in traditional schools.

Scholars and policy makers have much to learn about why charter middle schools in LA select a wide range of students, then appear to lift achievement higher, relative to their TPS counterparts. It's not inevitable that charters would differentially select differing families or students. Our findings do suggest that many charter schools will continue to draw out higher achieving students from traditional schools, and this may compromise the likelihood that TPS can adjust and compete by reforming from the inside.

Finally, evaluation researchers often endeavor to link variation in the internal features of organizations with varying results experienced by their students or clients. We have shown the utility of backing up to understand how segmented sets of schools are becoming more diverse in a lively and less regulated field of institutions. The contrasting features of these diverse organizations—each vying for stronger students, each advancing particular educational aims and perhaps class interests—then set a telling causal chain in motion. Researchers could better inform stakeholders and policy makers by capturing this entire process—illuminating how schools serve differing kids and families, acquire teachers and resources of varying quality, then yield unequal achievement effects.

Notes

1. The rigor of authorizing agencies may be another consequential element of the regulatory environment. But neither the count of authorized start-up and conversion charters, nor the selectivity of student admissions permitted under local policies, has been found to be predictive of achievement gains (Gleason et al., 2010; Zimmer et al., 2012).
2. Traditional schools may shed less effective teachers and raise compensation for more effective teachers when a charter school opens nearby, as Jackson (2012) found, drawing on North Carolina data. Cowen and Winters (2013) found higher turnover rates in Florida charter schools, compared with TPS, over the 2002–2008 period. Less effective teachers were more likely to exit than more effective teachers, but these rates did not differ significantly between sectors.
3. Focusing on the Milwaukee school voucher program, Fleming, Cowen, Witte, and Wolf (2015) found that parents drawing school vouchers earned less than TPS peers but the former displayed significantly higher levels of educational attainment.
4. In contrast, Hoxby and Murarka (2009) found no differences among ethnic groups for pupils attending New York City charter schools.
5. Most LAUSD students enter middle school at Grade 6 and high school at Grade 9.
6. Care must be taken in estimating a propensity score, because inclusion of too many variables, even though correlated with the treatment, can actually induce overt bias in the matched samples by reducing the overlaps between the treatment group and control group (Lesaffre & Albert, 1989).

7. The algorithm works by finding improvements in the most imbalanced variables, gradually improving balance over successive iterations during which each variable is weighed according to its relative importance for achieving the best balance. One may use the genetic algorithm by drawing from the propensity score and the covariates after they have been made orthogonal to it. If optimal balance is achieved by simply matching on the propensity score, then the other variables are given a zero weight and genetic matching will be equivalent to propensity score matching. One advantage of the genetic matching algorithm is that it directly optimizes covariate balance.
8. The Kolmogorov-Smirnov (KS) test is a nonparametric test for the equality of continuous, one-dimensional probability distributions that can be used to compare two samples, and the KS test is sensitive to imbalance across the empirical distribution.
9. Fewer differences for ESBMM schools emerged, except that the two high schools served higher shares of Asian students, perhaps another particular niche within this field of deregulated firms.

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Appendix. Details of model specifications

The identifications estimated for *switching* students and families are for those who switched schools at baseline in 2007–2008 and then attended the same school until the final year, 2010–2011. Identifications for *stayers* are for students who stayed in the same school from 2007–2008 to 2010–2011. For the OLS regressions, we consider two empirical models below building on the basic approach in Zimmer and Buddin (2006a):

Reg I :

$$Y_i = \alpha + \gamma \text{Treat}_i + \beta \text{Baseline}_i + \lambda X_i + \epsilon_i,$$

Reg II:

$$Y_{it} = \mu_i + \gamma \text{Treat}_{it} + \beta \text{Time}_{it} + \epsilon_{it}$$

.

These two OLS regressions have different data structures, Reg I views the data structure as single-level and Reg II views the data structure as two-level and uses student-level fixed-effects model. First, in Reg I, Y_i indicates outcome score in ELA or math for individual i in the final year, $Treat_i$ is a dummy variable which indicates if individual i attends a charter school. Thus, γ is a parameter of our interest reflecting the effect of charter school attendance on outcome variable. $Baseline_i$ is a test score of individual i in the baseline year and is considered as a key covariate in our identification strategy. $\mathbf{X}_i$ is a vector of student-level covariates including the child's ethnicity, designation as limited English proficient and special education eligible, the PED social-class proxy, and parent education (less than high school diploma, some college, or more).

In Reg II, similar to Reg I, Y_{it} is the test score in ELA or math for individual i at time t and $Treat_{it}$ is a dummy variable which indicates if individual i attends a charter school at time t . Because Reg II is a student-level fixed-effects model, student-level covariates cannot be specified as covariates. Instead, we use the $Time_{it}$ variable that reflects mean chronological change across each year controlling for the effects from treatment. μ_i indicates the student-level fixed effects term, and γ in Reg II yields the estimator controlling for any unobserved individual differences. Note again that student-level fixed-effects models cannot specify time-invariant covariates, and this does not allow us to estimate charter school effects for *stayers*, who do not change the status of charter school attendance across years. Moreover, because outcome measures used in this study are not vertically scaled across years, they are not directly comparable across years and differences between consecutive years do not have interpretable meaning. Thus, we chose not to pursue the “difference out” approach.

Modeling Selection into ‘Treatment’ Schools using Propensity Score Matching and Genetic Matching

For the propensity score matching analysis, we use probit regression that specifies baseline test score, special education status, limited English proficiency status, subsidized lunch social-class proxy, and categorized ethnicity and parent education as covariates. For parent education the “college or above” category is used as the reference group. We use GLM function and MATCH function in the software, R, to obtain propensity scores and the average treatment effect on the treated (ATT). [Appendix Table 1A](#) presents the results of the propensity score estimation for the example of standardized math scores for *switchers* attending middle school.

Appendix Table 1A. Factors affecting the likelihood of attending a charter school obtained from the probit regression.

	Coefficient	<i>p</i> -value	Standard error
Baseline	0.01		0.01
Disability/Special education	−0.11	*	0.04
English learner	−0.10	**	0.03
PED social-class proxy	−0.35	***	0.03
Asian	−0.24		0.21
Hispanic	−0.10		0.21
Black	0.27		0.21
White	0.34	.	0.21
Less than high school (parent)	0.02		0.03
High school graduate (parent)	−0.16	***	0.03
Number of observations	21,646		
Number of schools	130		
AIC	12560		

Note. AIC = Akaike information criterion.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

For the genetic matching analyses we use the same covariates that are used for obtaining the propensity scores. We run the GenMatch function in R using the default loss function, which implies “lexical” optimization: All of the balance statistics will be sorted from the most discrepant to the least and weights will be picked which minimize the maximum discrepancy. In solving the optimization problem, we use 1,000 population size, which seems large enough to find valid solutions. We also report average treatment effect on the treated (ATT).

Sensitivity Analysis of Genetic Matching Results

For the sensitivity analysis, we used Rosenbaum’s method (Keele, 2010). In this approach, different levels are set for Γ , the log odds of receiving treatment. The test then assesses the lower and upper bounds of a matching estimate when one observation in a matched pair is allowed to have a higher probability of receiving treatment due to observed or unobserved confounders. For example, when setting $\Gamma = 3$, one observation in a matched pair could be three times as likely to have received treatment without eliminating the observed effect of that treatment. If the bounds include zero at low levels of Γ , then the estimate should be considered highly sensitive to selection bias.

In our case the p -values of the effect for middle-school charter switchers after genetic matching are significant at the conventional level ($\alpha = 0.05$) until $\Gamma = 1.84$ for English language arts (ELA) and $\Gamma = 1.61$ for math. The corresponding values are 1.40 for elementary-school charter stayers in ELA 1.49 for high-school charter stayers in ELA.

These results suggest that middle school results are more robust for ELA and math gains than for charter-elementary benefits, or charter-high benefits. That is, elementary-level findings for ELA is more sensitive to potential bias introduced by unobserved confounders.

The same procedure helps to set bounds for interpreting differential achievement effects for start-up and conversion charter schools. We found that $\Gamma = 1.28$ for students in middle-school conversions for ELA result; $\Gamma = 1.71$ for pupils attending middle-school start-ups for math gains, respectively; and $\Gamma = 1.58$ and $\Gamma = 1.26$ for pupils attending high school start-ups and conversions for ELA gains, respectively. Again, we see the most robust effects in middle-school start-ups for both ELA and math gains. Yet, despite significant reduction in overt bias using the matching techniques, the risk of bias stemming from unobserved confounders remains, especially for benefits pertaining to conversion charter school attendees.

Affiliated versus independent conversion charter schools

We provide [Appendix Table 1B](#) for baseline score comparisons for affiliated versus nonaffiliated conversion startup charters in ELA and mathematics. This supports our findings that affiliated charters have higher baseline test scores than nonaffiliated charters in both subjects in conversions and startup charters.

Appendix Table 1B. Baseline score comparison across affiliated versus nonaffiliated for conversion charters and startup charters in English Language Arts (ELA) and mathematics.

	Conversion affiliated	Conversion nonaffiliated	Startup affiliated	Startup nonaffiliated
ELA	0.77	0.12	0.88	0.06
Math	0.60	0.16	0.76	0.14

Appendix Table 2. Results for middle school charter effects in English Language Arts (ELA), including student ethnicity interaction terms, by type of charter.

Covariate	Coef	SE	t	p-value
ELA—middle school conversion charters				
Baseline	0.72	0.00	140.85	0.00
Conversion	0.05	0.03	1.39	0.16
Dis1	−0.04	0.01	−3.15	0.00
Lang1	−0.13	0.01	−13.35	0.00
nslp1	−0.07	0.01	−6.78	0.00
White	0.19	0.07	2.59	0.01
Black	−0.07	0.07	−1.04	0.29
Hispanic	−0.01	0.07	−0.21	0.83
Asian	0.22	0.07	3.06	0.00
Wh_tr	−0.03	0.03	−1.18	0.23
Bl_tr	0.11	0.04	2.74	0.00
Hi_tr	0.22	0.01	12.79	0.00
As_tr	0.09	0.05	1.76	0.07
High	−0.04	0.00	−4.93	0.00
Bhigh	−0.02	0.01	−2.28	0.02
_Cons	0.15	0.07	2.07	0.03
ELA—middle school startup charters				
Baseline	0.72	0.00	140.87	0.00
Startup	−0.03	0.03	−1.03	0.30
Dis1	−0.04	0.01	−3.14	0.00
Lang1	−0.13	0.01	−13.34	0.00
nslp1	−0.08	0.011	−6.79	0.00
White	0.18	0.07	2.51	0.01
Black	−0.08	0.07	−1.12	0.26
Hispanic	−0.02	0.07	−0.28	0.77
Asian	0.22	0.07	2.98	0.00
Wh_tr	0.00	0.03	0.11	0.91
Bl_tr	0.15	0.05	2.93	0.00
Hi_tr	0.26	0.03	6.95	0.00
As_tr	0.13	0.05	2.26	0.02
High	−0.04	0.00	−4.93	0.00
Bhigh	−0.02	0.01	−2.29	0.02
_Cons	0.15	0.07	2.15	0.03